DETECTION OF FAKE AND AI-GENERATED PRODUCT REVIEWS USING DEEP LEARNING: A COMPARATIVE STUDY OF FOUR ARCHITECTURE FAMILIES

Synopsis report on
Deep Learning Project
[ICT-4442]

Submitted By

Rudra Rajpurohit (Reg No. 230911570)
Vedant Totla (Reg No. 230911526)
Ishan Satyanand Thakur (Reg No. 230953356)
Vedant Agarwal (Reg No. 230953312)



SCHOOL OF COMPUTER ENGINEERING
MANIPAL INSTITUTE OF TECHNOLOGY,
MANIPAL ACADEMY OF HIGHER EDUCATION

OCTOBER 2026

PART A : SYNOPSIS (Phase 1)

Team No. and Names (with Registration Numbers): Team 14

S. No.	Name	Reg. No.	Branch / Section
1	Rudra Rajpurohit	230911570	IT / B
2	Vedant Totla	230911526	IT / B
3	Ishan Satyanand Thakur	230953356	CCE / C
4	Vedant Agarwal	230953312	CCE / C

Title of the Project: Detection of Fake and AI-Generated Product Reviews Using Deep Learning: A Comparative Study of Four Architecture Families

Brief Description of the Project (not more than 500 words):

Online product reviews guide purchases only as long as customers wrote them. Review farms have broken that assumption for years, and large language models now break it cheaply by writing fluent, product-specific reviews on demand. Salminen et al. [1] found that human raters identify GPT-2 reviews with 55.36% accuracy, barely above chance. Platforms therefore have to detect such reviews automatically.

This project treats detection as binary text classification and compares four deep learning architecture families under identical conditions. The primary dataset is the Fake Reviews Dataset of Salminen et al. [1], which pairs about 20,000 GPT-2 reviews with 20,000 original Amazon reviews across ten product categories. The Deceptive Opinion Spam corpus of Ott et al. [2], [3] is used only as a cross-domain test set.

The four models cover the families taught in the course. TF-IDF + MLP is a feed-forward baseline on word and character n-grams that shows how much of the task lexical statistics alone can solve. Text-CNN applies filters of widths 3, 4 and 5 over word embeddings to catch local phrase patterns such as stock praise. BiLSTM + Attention reads each review in both directions and pools its hidden states with attention, which also shows the tokens behind each prediction. DistilBERT is a pre-trained Transformer encoder fine-tuned for the task, from the family that holds the best published results.

All four models share one preprocessing pipeline, one stratified 70/15/15 split with a fixed seed and one evaluation script for accuracy, precision, recall, macro F1 and ROC-AUC. Parameter count, training time and inference latency are recorded beside accuracy, so the comparison also covers deployment cost. The deliverables are a single comparison table, an error analysis of shared failures and an explanation of where the architectures differ, in particular whether pre-trained contextual representations are needed or shallower models suffice.

Dataset(s) Identified (with details of the dataset):

Primary: the Fake Reviews Dataset of Salminen et al. [1], available on OSF and Kaggle. The Kaggle copy holds 40,526 labelled reviews, 20,294 computer-generated by a GPT-2 model trained on Amazon reviews and 20,232 original Amazon reviews. Each review has a product category, a star rating, the text and a label. The ten main categories include Books, Clothing, Electronics, Home and Kitchen, and Toys and Games. The classes are balanced, so accuracy and macro F1 are both meaningful.

Secondary, for the cross-domain test: the Deceptive Opinion Spam Corpus v1.4 of Ott et al. [2], [3]. It holds 1,600 reviews of 20 Chicago hotels, 800 truthful reviews from TripAdvisor and other review sites and 800 deceptive reviews written by Amazon Mechanical Turk workers, balanced across positive and negative sentiment. It is never used for training.

Common protocol: one stratified 70/15/15 split of the primary dataset with random seed 42, identical for every model.

Proposed Models for Comparison (3–4, spanning different architecture families, e.g., a classical/MLP baseline, a CNN variant, an RNN/LSTM variant, or an attention-based model):

Sl. No.	Model	Architecture family	Rationale
1	TF-IDF + MLP	Feed-forward	Lower bound: how far word and character statistics go without word order
2	Text-CNN (filter widths 3, 4, 5)	Convolutional	Captures local phrase patterns such as stock praise at low training cost
3	BiLSTM + Attention	Recurrent	Models word order and longer context, and its attention weights show which tokens drive each prediction
4	DistilBERT (fine-tuned)	Transformer, self-attention	Pre-trained contextual representations, the family behind the best published results

Model-to-Member Assignment (one model/task per member):

Member	Reg. No.	Model owned	Shared responsibility
Rudra Rajpurohit	230911570	DistilBERT	Evaluation harness, metric scripts and comparison table
Vedant Totla	230911526	BiLSTM + Attention	Cross-domain transfer test and error analysis
Ishan Satyanand Thakur	230953356	Text-CNN	Preprocessing and tokenisation pipeline, embedding setup
Vedant Agarwal	230953312	TF-IDF + MLP	Dataset acquisition, exploratory analysis, literature review table

Existing State-of-the-Art Method(s) Related to the Proposed Project:

Ott et al. [2] built the standard deceptive opinion spam benchmark and reached 89.8% accuracy with a linear SVM on psycholinguistic and bigram features, while untrained human judges scored between 53% and 62%.

Ren and Zhang [4] combined CNN sentence vectors with a gated recurrent network with attention and outperformed feature-based methods on hotel, restaurant and doctor reviews. Recurrent models with attention are therefore a strong reference point.

Salminen et al. [1] generated fake reviews with GPT-2 and detected them with a fine-tuned RoBERTa at 97% precision and recall, far above human raters at 55.36% accuracy. This is the reference result the project targets with a lighter Transformer.

DetectGPT [5] detects machine-generated text without labelled data and reached 0.95 AUROC on GPT-NeoX news text. It needs the log probabilities of the generating model, which a review platform does not have.

Objectives (crisp and clear, not more than 2–3; must not describe activities/steps):

1) To quantify the detection performance of four architecture families, namely feed-forward, convolutional, recurrent and self-attention, on fake and AI-generated product reviews under one common evaluation protocol.
2) To characterise the trade-off between detection accuracy and computational cost, measured as parameter count, training time and inference latency, for each of the four architectures.
3) To determine how far detectors trained on machine-generated product reviews keep their accuracy on human-written deceptive reviews from a different domain.

Importance / Motivation of the Proposed Project:

Review fraud has a direct economic cost. Inflated ratings steer spending toward worse products, penalise honest sellers and erode trust in a platform once buyers realise they were misled. Regulators have started to act. In August 2024 the US Federal Trade Commission issued a final rule banning fake consumer reviews and testimonials [6], which puts pressure on platforms to find them at scale. The technical problem has changed too. Fakes written by language models are fluent, specific and cheap to produce in volume, and human raters cannot reliably identify them [1], so manual moderation does not scale. For the course, the task suits a controlled architecture comparison. The inputs are short texts, the labels are reliable and the classes are balanced, and every architecture family in the syllabus has a natural formulation on it. Differences in results can then be traced to the architecture and not to the data.

Preliminary Reference List (minimum 5 papers):

[1] J. Salminen, C. Kandpal, A. M. Kamel, S. Jung and B. J. Jansen, "Creating and detecting fake reviews of online products," J. Retailing and Consumer Services, vol. 64, art. 102771, 2022.

[2] M. Ott, Y. Choi, C. Cardie and J. T. Hancock, "Finding deceptive opinion spam by any stretch of the imagination," in Proc. 49th Annu. Meeting Assoc. Comput. Linguistics (ACL-HLT), Portland, OR, USA, 2011, pp. 309-319.

[3] M. Ott, C. Cardie and J. T. Hancock, "Negative deceptive opinion spam," in Proc. Conf. North American Chapter Assoc. Comput. Linguistics (NAACL-HLT), Atlanta, GA, USA, 2013, pp. 497-501.

[4] Y. Ren and Y. Zhang, "Deceptive opinion spam detection using neural network," in Proc. 26th Int. Conf. Computational Linguistics (COLING), Osaka, Japan, 2016, pp. 140-150.

[5] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning and C. Finn, "DetectGPT: Zero-shot machine-generated text detection using probability curvature," in Proc. 40th Int. Conf. Machine Learning (ICML), Honolulu, HI, USA, 2023.

[6] Federal Trade Commission, "Federal Trade Commission announces final rule banning fake reviews and testimonials," press release, Aug. 2024. [Online]. Available: https://www.ftc.gov/news-events/news/press-releases/2024/08/federal-trade-commission-announces-final-rule-banning-fake-reviews-testimonials

[7] Y. Kim, "Convolutional neural networks for sentence classification," in Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP), Doha, Qatar, 2014, pp. 1746-1751.

[8] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997.

[9] A. Vaswani et al., "Attention is all you need," in Advances in Neural Information Processing Systems (NIPS), Long Beach, CA, USA, 2017, pp. 5998-6008.

[10] J. Devlin, M.-W. Chang, K. Lee and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, Minneapolis, MN, USA, 2019, pp. 4171-4186.

[11] V. Sanh, L. Debut, J. Chaumond and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," arXiv:1910.01108, 2019.

[12] N. Jindal and B. Liu, "Opinion spam and analysis," in Proc. Int. Conf. Web Search and Data Mining (WSDM), Palo Alto, CA, USA, 2008, pp. 219-230.

Work Plan:

Sl. No.	Component / Work Element / Milestone	Expected Completion (Day/Month)
1	Team formation, problem finalisation and synopsis submission	25 Aug 2026
2	Dataset acquisition, exploratory data analysis and class balance check	01 Sep 2026
3	Shared preprocessing pipeline (cleaning, tokenisation, encoding) and fixed 70/15/15 stratified split	08 Sep 2026
4	Common evaluation harness with accuracy, precision, recall, macro F1 and ROC-AUC	12 Sep 2026
5	Literature review of 8 to 10 papers compiled into the summary table	18 Sep 2026
6	TF-IDF + MLP and Text-CNN implemented with preliminary results	22 Sep 2026
7	Interim report and individual contribution log submitted	25 Sep 2026
8	BiLSTM + Attention implemented and evaluated	06 Oct 2026
9	DistilBERT fine-tuning completed and evaluated	13 Oct 2026
10	Hyperparameter tuning, cross-domain transfer test and error analysis	20 Oct 2026
11	Comparison table, final report draft, plagiarism and AI-content check, slides	25 Oct 2026
12	In-class presentation and viva	26 to 29 Oct 2026
13	Final report and GitHub repository link submitted	31 Oct 2026

Use of AI tools. Claude (Anthropic) was used to help draft parts of this synopsis.